THAVATHIRU SANTHALINGA ADIGALAR ARTS,SCIENCE &TAMIL COLLEGE
PERUR-COIMBATORE

SAFE GAURD

DOMAIN-WOMEN'SSAFETY&SOCIAL SECURITY
"CYBER SAFETY AWARENESS/ PHISHING DETECTION PLATFORM"

PRESENTED BY:

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PROBLEM OVERVIEW

Women face:

- Physical threats in public spaces
- Harassment (online & offline)
- Cyber threats like phishing and scams
- Delayed emergency response

Existing systems:

- Require manual dialing
- Do not ensure anonymity
- Lack integrated digital safety tools
- Target Users are Women, Students, Working professionals, Harassment victims, Cyber fraud victims

SOLUTION OVERVIEW

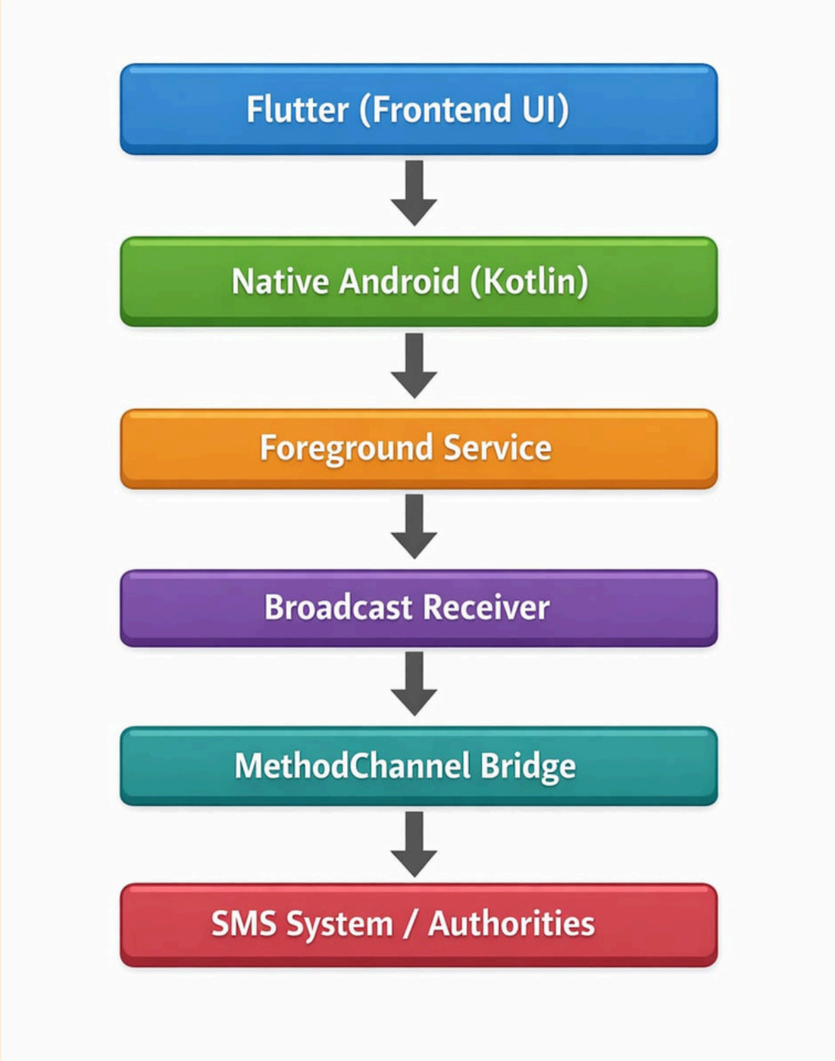
Safe Gaurd is a mobile-based intelligent safety system that provides:

- Emergency SOS via Power Button (3 presses)
- Anonymous Harassment Reporting
- Phishing Risk Detection (Next Module)

Key Capabilities

- Background SOS detection
- Live GPS location sharing
- Anonymous complaint submission
- Consent-based contact sharing
- Android 15 compatible architecture
- Foreground service monitoring

TECHNICAL ARCHITECTURE



PROTOTYPE OVERVIEW

Demonstrate automatic SOS triggering using the power button

- Validate real-time GPS location sharing
- Test anonymous harassment reporting system
- Verify the hybrid Flutter + Native Android architecture

This prototype ensures that the system works reliably on modern Android versions including Android 15.

PROTOTYPE DEMONSTRATION

SOS Emergency System

- Power button pressed 3 times
- Foreground service detects screen toggles
- Signal sent via MethodChannel
- GPS location retrieved
- Google Maps link added to SMS

Harassment Reporting

- User writes complaint
- Anonymous or consent-based reporting
- Complaint converted to SMS
- Sent to authorities

My Case Tracking

- Case ID – unique identifier
- Type of Case – harassment / safety
- Severity – low / medium / high
- Status – pending / in progress / resolved

MVP Features

- SOS Power Button Trigger
- Live GPS Location Sharing
- Emergency SMS Alert
- Anonymous Reporting
- Case Tracking System

SOS EMERGENCY WORKFLOW

Power Button Press (3 Times)
↓
Screen ON/OFF Events Detected
↓
Foreground Service Monitors Screen Activity
↓
Emergency Trigger Condition Verified
↓
Native Android Layer Sends Signal via MethodChannel
↓
Flutter Application Receives SOS Event
↓
User's GPS Location Retrieved
↓
Location Converted to Google Maps Link
↓
Emergency Message Generated
↓
SMS App Opens with Pre-filled Alert Message
↓
User Sends Message to Emergency Contacts

HARASSMENT REPORT WORKFLOW

User Opens Complaint Module
↓
User Writes Complaint Description
↓
User Selects Privacy Option
• Anonymous Reporting
• Consent-Based Contact Sharing
↓
Complaint Message Generated
↓
Complaint Converted to SMS Format
↓
SMS Prepared for Authorities
↓
User Sends Complaint

PROGRESS

UI Development (Flutter):

- Built using Material Design components
- Implemented MaterialApp, Scaffold, AppBar
- Added ElevatedButton, TextField for user actions
- Integrated Navigator routing for screen navigation

Backend Setup (FastAPI):

- Developed Emergency Alert Mode
- Implemented Harassment Reporting Module
- Created Information Storage System for reports

Module Integration:

- Connected UI with backend services
- Enabled report data handling and storage

FURTHER STEP

AI Phishing Detection

- Implement AI model to analyze suspicious links
- Classify threats as High, Medium, or Low Risk

Threat Analytics

- Develop dashboard to visualize safety statistics
- Track total reports and risk-level cases

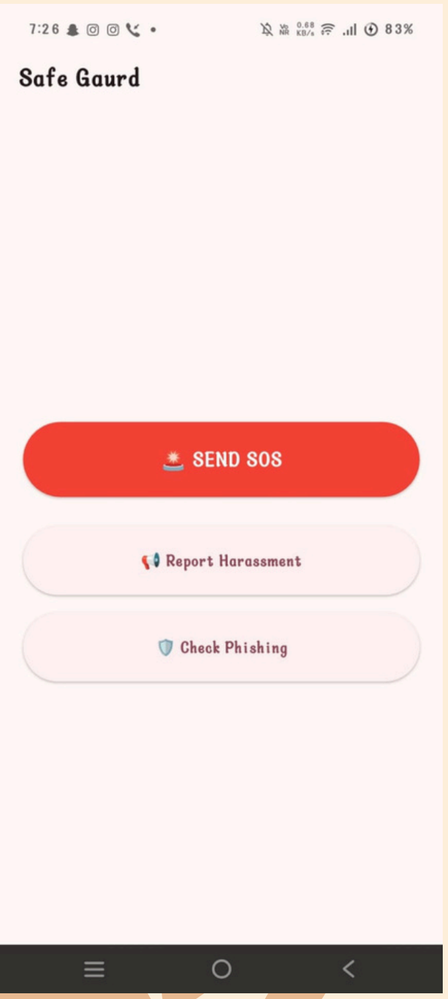
My Cases Module

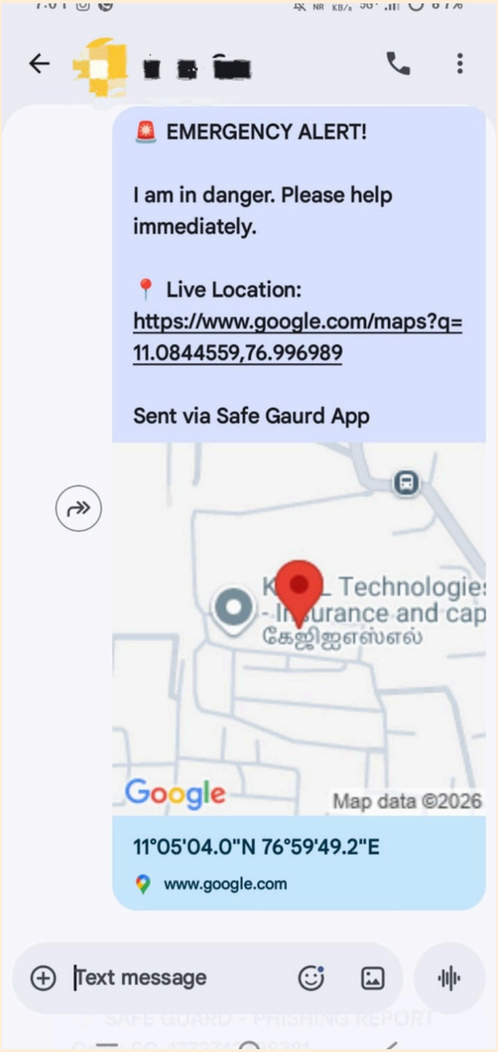
- Allow users to track submitted reports
- View status and history of anonymous complaints

Testing & Optimization

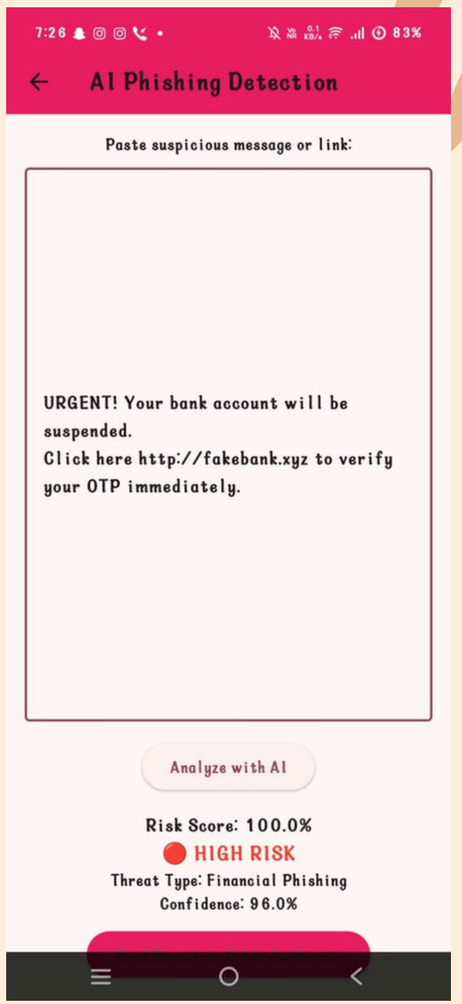
- Improve system performance and reliability
- Complete integration between frontend and backend

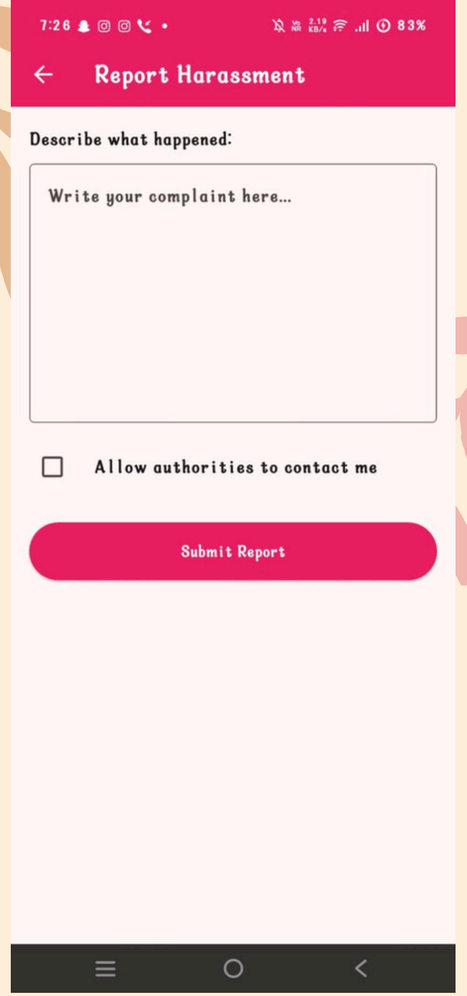
OUTPUT


7:26
83%
Safe Gaurd
SEND SOS
Report Harassment
Check Phishing


EMERGENCY ALERT!
I am in danger. Please help immediately.
Live Location:
https://www.google.com/maps?q=11.0844559,76.996989
Sent via Safe Gaurd App
Google
Map data ©2026
11°05'04.0"N 76°59'49.2"E
www.google.com
Text message

7:26
83%
AI Phishing Detection
Paste suspicious message or link:
Paste message here...
Analyze with AI
English


7:26
83%
AI Phishing Detection
Paste suspicious message or link:
URGENT! Your bank account will be suspended.
Click here http://fakebank.xyz to verify your OTP immediately.
Analyze with AI
Risk Score: 100.0%
HIGH RISK
Threat Type: Financial Phishing
Confidence: 96.0%


7:26
83%
Report Harassment
Describe what happened:
Write your complaint here...
Allow authorities to contact me
Submit Report

Thank You